

A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form

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Abstract

The Connes–van Suijlekom and Connes–Consani–Moscovici truncations of the Weil quadratic form, at a prime cutoff $c > 1$ and frequency band N , produce finite Galerkin matrices whose spectra are the finite-rank window on Weil positivity. We prove two exact finite theorems about this truncation. First, every real even Galerkin coefficient vector v determines, in closed form, a band-limited Guinand–Weil test function g_v with

$$\langle v, Q_\infty v \rangle = \sum_{\zeta(1/2+iz)=0}^* g_v(z),$$

zeros counted with multiplicity, Q_∞ the cutoff-free truncated matrix: every value of the truncated form is an exact sum over the nontrivial zeros of ζ . The construction factors through an exact source quotient of dimension $2N + 1$ and admits a non-collapsing pole-neutral subfamily. Second, beyond the Galerkin band the omitted archimedean tail is a totally positive Cauchy–Stieltjes increment. This yields a two-sided certification rule with an explicit budget $B_T \sim (2N+1)\rho \log T/(\pi^2 T)$, where T is the archimedean cutoff and $\rho = 2\pi/\log c$: finite-cutoff positivity certifies cutoff-free positivity, a finite-cutoff eigenvalue below $-B_T$ certifies a cutoff-free negative, and a negative eigenvalue in the band $[-B_T, 0)$ certifies nothing. Resolving a spectral scale of 10^{-59} at $c = 100$ by brute cutoff would require T of order 10^{63} ; a cutoff-free interval LDL^T factorization resolves it directly. The dictionary is verified over the first 512 zeros of ζ and by three independent computational routes; all scripts and artifacts ship with the paper.

1 Introduction

The Riemann–Weil explicit formula relates the nontrivial zeros of $\zeta(s)$ to prime, pole, and archimedean terms, and Weil positivity, the positivity of the associated quadratic form on a suitable class of test functions, is equivalent to the Riemann hypothesis [28, 19, 4, 29]. In Connes’s program the Weil form is the central object of a spectral attack on the zeros [6, 8, 9, 7], and recent work of Connes–van Suijlekom [11] and Connes–Consani–Moscovici [10] turns the form into concrete finite objects: for a prime cutoff $c > 1$ and a band N , an explicit $(2N+1) \times (2N+1)$ Galerkin matrix whose deep spectrum is the finite-rank window on Weil positivity. Suzuki’s screw-function framework [26] gives a parallel continuous-variable treatment of the same form.

This note isolates the part of the truncated structure that is exact at every finite level, and proves a finite-cutoff order theorem that governs what finite numerics can and cannot certify about it. The two results are independent theorems about the same object, and together they turn the truncated matrix into a calibrated instrument: its values are exact zero sums, and its finite-cutoff spectra carry explicit certification bounds.

First result. A real even coefficient vector $v \in \mathbb{R}^{N+1}$ canonically determines a chain

$$v \longmapsto T_v \longmapsto K_v \longmapsto \widehat{g}_v \longmapsto g_v,$$

ending in an entire, band-limited Guinand–Weil test function g_v , and for this induced test function the cutoff-free truncated form evaluates the zero side of the explicit formula exactly (Theorem 2.5): *every entry of the Connes–van Suijlekom/Connes–Consani–Moscovici matrix, and every quadratic value $\langle v, Q_\infty v \rangle$, is an exact finite-form sum over the nontrivial zeros of ζ .* The same calculation shows that all finite signed source distributions act through an exact quotient of dimension $2N + 1$ (Corollary 2.4), and that the pole term can be switched off on a non-collapsing subfamily of dimension $N - s - 1$ (Corollary 2.7), which is the finite analogue of working orthogonally to the pole in Weil’s criterion. No limit in N and no archimedean cutoff enters the dictionary: the coefficient vector is transported to one compactly supported test function before any numerical quadrature exists.

Second result. For the archimedean term, finite computations must truncate an integral at some cutoff T . Theorem 3.2 proves that past the Galerkin band the omitted tail is a rank-two-density Cauchy–Stieltjes Gram increment: for $T_2 > T_1 > \max(\rho N, 7)$ the matrix increment $Q_{\text{arch}, T_2} - Q_{\text{arch}, T_1}$ is positive definite, and in the natural frequency order it is strictly totally positive. The consequence (Corollary 3.3) is a two-sided decision rule with an explicit, elementary budget B_T :

$$\lambda_j(Q_T^{\text{tot}}) < \lambda_j(Q_\infty) \leq \lambda_j(Q_T^{\text{tot}}) + B_T, \quad B_T = \frac{(2N+1)\rho}{\pi^2 T} \left(\log \frac{T}{2\pi} + 1 \right) (1 + o(1)).$$

Finite- T positivity therefore certifies cutoff-free positivity, a finite- T eigenvalue below $-B_T$ certifies a cutoff-free negative, and a negative eigenvalue inside the band $[-B_T, 0)$ certifies nothing. Since B_T decays only like $\log T/T$, each doubling of T buys about one binary digit: deep spectral scales are unreachable by brute cutoff, and the correct finite instrument at depth is the cutoff-free assembly itself.

The second theorem is motivated by a correction to an earlier unpublished finite-cutoff computation: at $(c, N, T) = (100, 200, 800)$ a fixed archimedean cutoff produced deep negative eigenvalues that are controlled by the omitted tail, not by the cutoff-free form. Andrews’s independent reproduction and Silva’s quadrature-sensitivity and exact-entry analyses made the cutoff-free treatment explicit [2, 16, 12]; the tail-order theorem is the finite algebraic rule behind that phenomenology, and Section 3 quantifies it: certifying the sign of a 10^{-59} -scale eigenvalue at $c = 100$ through the cutoff would require $T \approx 8 \cdot 10^{62}$, while a cutoff-free interval $LDL^\top$ factorization of the same matrix certifies $n_- = 0$ directly.

Relation to prior work. The zero-sum identity of Theorem 2.5 is the Weil explicit formula applied to the induced test function; the autocorrelation mechanism $g = f * \tilde{f}$ goes back to Weil [28], finite-dimensional restrictions of the Weil form were studied by Yoshida [29], and the divided-difference structure of the truncated matrix is Proposition 4.1 of Connes–van Suijlekom [11]. The contribution here is the exact closed-form transport: the explicit chain $v \mapsto g_v$ with its inverse-free calculus (Lemma 2.3), the identification of the assembled matrix with the Connes–Consani–Moscovici closed forms at entry level (Lemma 2.1), the exact $2N+1$ source quotient, the pole-neutral family, and the tail-order theorem with its strict total positivity and explicit budget. Band-limited test functions are a standard device in explicit-formula analysis (see for instance [5] and the literature cited there); what is used here is the exact match between that class and the finite Connes–van Suijlekom/Connes–Consani–Moscovici data. The total-positivity input is classical Cauchy-kernel theory [21, 25, 3], and Silva records adjacent Loewner, Herglotz, and operator-monotone structure for the prime and pole pieces [13, 14, 15, 17].

All numerical statements in this paper are backed by a released verification package (Section 4) containing exact symbolic audits, interval-arithmetic certificates computed with Arb [20], and a three-route confirmation of the dictionary identity, including a sum over the first 512 zeros of ζ .

2 The finite dictionary

2.1 From coefficient vectors to test functions

Fix $c > 1$, put

$$L = \log c, \quad \Delta = \frac{L}{2\pi}, \quad \rho = \frac{2\pi}{L},$$

and fix $N \geq 0$. A real even-sector Galerkin vector $v = (v_0, \dots, v_N)$ is embedded into symmetric full coefficients by

$$u_0 = v_0, \quad u_k = u_{-k} = \frac{v_k}{\sqrt{2}} \quad (1 \leq k \leq N),$$

which is the isometric even-sector embedding used by Connes–van Suijlekom and Connes–Consani–Moscovici. Let

$$T_v(t) = \sum_{m=-N}^N u_m e^{2\pi i m t}$$

and define the Volterra sine-chord kernel

$$K_v(\omega) = 2 \int_0^\omega T_v(t) T_v(\omega - t) dt.$$

The induced Fourier weight and test function are

$$\widehat{g}_v(\xi) = \begin{cases} \pi K_v(1 - |\xi|/\Delta), & |\xi| \leq \Delta, \\ 0, & |\xi| > \Delta, \end{cases} \quad g_v(z) = \int_{-\Delta}^{\Delta} \widehat{g}_v(\xi) e^{2\pi i z \xi} d\xi.$$

The weight is even, hence g_v is even.

For a C^1 source function ψ on $\mathbb{R}$, let Q_ψ denote the finite divided-difference matrix on indices $I_N = \{-N, \dots, N\}$:

$$(Q_\psi)_{mn} = \begin{cases} \frac{\psi(m) - \psi(n)}{m - n}, & m \neq n, \\ \psi'(m), & m = n, \end{cases}$$

exactly the structure of [11, Prop. 4.1], and write

$$\langle v, Q_\psi v \rangle = \sum_{m,n=-N}^N u_m u_n (Q_\psi)_{mn}$$

for the even-sector contraction. The three source functions of the truncated Weil form are

$$\psi_p^{(c)}(x) = -\frac{1}{\pi} \sum_{q=p^a \leq c} \frac{\Lambda(q)}{\sqrt{q}} \sin\left(2\pi x \left(1 - \frac{\log q}{L}\right)\right), \quad (1)$$

$$\psi_0(x) = \frac{1}{\pi} \int_0^L 2 \cosh(y/2) \sin\left(2\pi x \left(1 - \frac{y}{L}\right)\right) dy, \quad (2)$$

$$\psi_{\mathbb{R},T}(x) = \frac{1}{2\pi^2} \int_{-T}^T h_+(r) \mathcal{S}(r, x, L) dr, \quad \mathcal{S}(r, x, L) = \int_0^L \sin\left(2\pi x \left(1 - \frac{y}{L}\right)\right) \cos(ry) dy, \quad (3)$$

where throughout

$$h_+(r) = \operatorname{Re} \psi_\Gamma\left(\frac{1}{4} + \frac{ir}{2}\right) - \log \pi.$$

We set

$$Q_{\text{prime}}^{(c)} := Q_{\psi_p^{(c)}}, \quad Q_{\text{pole}} := Q_{\psi_0}, \quad Q_{\text{arch},T} := Q_{\psi_{\mathbb{R},T}}, \quad Q_{\text{arch},\infty} := \lim_{T \rightarrow \infty} Q_{\text{arch},T},$$

the last limit existing entrywise (Lemma 2.1), and define the cutoff-free truncated Weil matrix

$$Q_\infty := Q_{\text{prime}}^{(c)} + Q_{\text{pole}} + Q_{\text{arch},\infty}. \quad (4)$$

Lemma 2.1 (Entry identification). *The limit defining $Q_{\text{arch},\infty}$ exists entrywise, and Q_∞ is the Connes–Consani–Moscovici Galerkin matrix of the Weil quadratic form: in the notation of [10, Eqs. (3.10)–(3.11), (3.16)],*

$$\langle v, Q_\infty v \rangle = W_{0,2}(F_v) - W_{\mathbb{R}}(F_v) - W_p(F_v), \quad F_v(x) = q(f_v, f_v)(\log x),$$

where f_v is the trigonometric polynomial with coefficient vector u on the interval of length L . In particular $-Q_{\text{prime}}^{(c)}$, Q_{pole} , and $-Q_{\text{arch},\infty}$ are the prime, pole, and archimedean blocks of the Connes–Consani–Moscovici assembly.

Proof. Each block is the divided-difference matrix of its source in the sense of [11, Prop. 4.1]. For the prime block, (1) is the source of the non-archimedean functional [10, Eq. (3.16)], so the identification is definitional. For the pole block, direct evaluation of (2) (as in Corollary 2.7 below) gives $\psi_0(n) = C_c n / (n^2 + \beta^2)$ and $\psi'_0(n) = C_c (\beta^2 - n^2) / (n^2 + \beta^2)^2$ with $\beta = L / (4\pi)$ and $C_c = L(\sqrt{c} + 1/\sqrt{c} - 2) / (2\pi^2)$, so that, using $\beta^2 = L^2 / (16\pi^2)$ and $16\pi^2 C_c = 8L(\sqrt{c} + 1/\sqrt{c} - 2) = 32L \sinh^2(L/4)$,

$$(Q_{\text{pole}})_{mn} = C_c \frac{\beta^2 - mn}{(m^2 + \beta^2)(n^2 + \beta^2)} = \frac{32L \sinh^2(L/4) (L^2 - 16\pi^2 mn)}{(L^2 + 16\pi^2 m^2)(L^2 + 16\pi^2 n^2)},$$

which is the pole matrix of [10, Lemma 4.1] verbatim. For the archimedean block, h_+ is the standard archimedean density of the explicit formula for the completed zeta function, i.e. the density of $W_{\mathbb{R}}$ in [10, Sec. 3] and [8, Eq. (153)]; the entrywise T -limit exists because at integer nodes $\mathcal{S}(r, n, L) = O(r^{-2})$ and $\partial_x \mathcal{S}(r, x, L)|_{x=n} = O(r^{-2})$ uniformly on I_N (both are computed in closed form in the proof of Theorem 3.2) while $h_+(r) = O(\log r)$. The released package additionally verifies the identification numerically: the closed-form Connes–Consani–Moscovici assembly and the source assembly (4) agree on generic (not pole- or moment-neutral) vectors to the working tail bound, e.g. to $2.0 \cdot 10^{-10}$ at $(c, N) = (29, 6)$ and to $2.1 \cdot 10^{-15}$ at $(c, N) = (13, 4)$ (Section 4). $\square$

Lemma 2.2 (Admissibility). *For every finite vector v , the induced function g_v is an admissible Guinand–Weil test function: it is entire of exponential type at most L , its Fourier transform is compactly supported in $[-\Delta, \Delta]$, and*

$$g_v(z) = O((1 + |\operatorname{Re} z|)^{-2})$$

uniformly on every fixed horizontal strip. Consequently g_v lies in the class for which the Guinand–Weil explicit formula holds with absolutely convergent zero sum [4, Sec. 2]: even, entire of finite exponential type, $\widehat{g}_v$ continuous of compact support, and decay $O((1 + |z|)^{-1-\delta})$ on horizontal strips with $\delta = 1$.

Proof. The Volterra convolution of finite exponential polynomials is entire in ω . Hence $\widehat{g}_v$ is compactly supported, continuous, piecewise smooth, and vanishes at $\pm\Delta$; the zero extension of $(\widehat{g}_v)'$ has bounded variation. The Paley–Wiener bound and the displayed decay follow by two integrations by parts, the second in the Stieltjes sense [24, 22]. The Riemann–von Mangoldt local zero count $N(t+1) - N(t) = O(\log t)$ then gives absolute convergence of the zero sum [27, Ch. 9]. $\square$

Lemma 2.3 (Finite source calculus). *Let*

$$\psi_{\alpha,\omega}(x) = \frac{\alpha}{\pi} \sin(2\pi\omega x).$$

For the finite divided-difference matrix attached to this source,

$$\langle v, Q_{\alpha,\omega} v \rangle = \alpha K_v(\omega).$$

Consequently, if μ is a finite signed Borel measure on $[0, 1]$ and

$$\psi_\mu(x) = \frac{1}{\pi} \int_{[0,1]} \sin(2\pi\omega x) d\mu(\omega),$$

then

$$\langle v, Q_\mu v \rangle = \int_{[0,1]} K_v(\omega) d\mu(\omega).$$

Proof. For $m \neq n$,

$$(Q_{\alpha,\omega})_{mn} = \frac{\alpha(\sin(2\pi\omega m) - \sin(2\pi\omega n))}{\pi(m - n)},$$

and the diagonal entry is

$$(Q_{\alpha,\omega})_{mm} = 2\alpha\omega \cos(2\pi\omega m).$$

On the other hand

$$K_v(\omega) = 2 \sum_{m,n=-N}^N u_m u_n e^{2\pi i n \omega} \int_0^\omega e^{2\pi i(m-n)t} dt.$$

For $m \neq n$ the real part of the summand is

$$u_m u_n \frac{\sin(2\pi\omega m) - \sin(2\pi\omega n)}{\pi(m - n)},$$

while for $m = n$ it is $2u_m^2 \omega \cos(2\pi\omega m)$. The imaginary parts cancel in pairs under the even symmetry $u_{-k} = u_k$. Multiplying by α gives the identity. For a finite signed measure, each single-frequency entry is continuous and bounded on $[0, 1]$; on the diagonal,

$$\psi'_\mu(x) = 2 \int_{[0,1]} \omega \cos(2\pi\omega x) d\mu(\omega).$$

The finite double sum may therefore be interchanged with the source integral. $\square$

Corollary 2.4 (Finite source quotient). *At level N , the finite source-to-form map $\mu \mapsto Q_\mu$ factors exactly through the $2N + 1$ coordinates*

$$\int \omega d\mu, \quad \int \sin(2\pi k \omega) d\mu, \quad \int \omega \cos(2\pi k \omega) d\mu \quad (1 \leq k \leq N),$$

and the induced map on these coordinates is injective: two finite signed sources produce the same level- N matrix if and only if all $2N + 1$ coordinates agree.

Proof. After polarization, the Volterra kernels are generated by $K_{1,1} = 2\omega$, $K_{1,\cos(2\pi kt)} = \sin(2\pi k\omega)/(\pi k)$, and

$$K_{\cos(2\pi kt),\cos(2\pi kt)} = \omega \cos(2\pi k\omega) + \frac{\sin(2\pi k\omega)}{2\pi k};$$

product-to-sum gives the mixed cases, and only frequencies $0, 1, \dots, N$ occur. Linear independence follows from the exponential-polynomial family $e^{2\pi i k\omega}$ and $\omega e^{2\pi i k\omega}$. Inspecting the entries $(0, 0)$, $(k, 0)$, and (k, k) recovers the displayed coordinates, so the quotient is exact in both directions. $\square$

2.2 The dictionary theorem

Theorem 2.5 (Finite Guinand–Weil dictionary). *For fixed $c > 1$, $N \geq 0$, and every real even finite Galerkin vector $v \in \mathbb{R}^{N+1}$,*

$$\langle v, Q_\infty v \rangle = \sum_{z \in Z_\zeta^*} g_v(z), \quad Z_\zeta^* = \{z \in \mathbb{C} : 1/2 + iz \text{ is a nontrivial zero of } \zeta\},$$

with multiplicity. Equivalently,

$$\langle v, Q_\infty v \rangle = -\frac{1}{\pi} \sum_{q=p^a \leq c} \frac{\Lambda(q)}{\sqrt{q}} \widehat{g}_v\left(\frac{\log q}{2\pi}\right) + 2g_v(i/2) + \frac{1}{2\pi} \int_{\mathbb{R}} h_+(r) g_v(r) dr.$$

Proof. Lemma 2.3 applied to the prime source (1), whose measure places mass $-\Lambda(q)/\sqrt{q}$ at $\omega_q = 1 - \log q/L$, gives

$$\langle v, Q_{\text{prime}}^{(c)} v \rangle = -\frac{1}{\pi} \sum_{q \leq c} \frac{\Lambda(q)}{\sqrt{q}} \pi K_v(\omega_q) = -\frac{1}{\pi} \sum_{q \leq c} \frac{\Lambda(q)}{\sqrt{q}} \widehat{g}_v\left(\frac{\log q}{2\pi}\right),$$

using $\widehat{g}_v(\xi) = \pi K_v(1 - \xi/\Delta)$ and $\log q/(2\pi) = \Delta(1 - \omega_q)$. Applied to the pole source, whose measure is the push-forward of $2 \cosh(y/2) L dy$ under $\omega = 1 - y/L$, it gives

$$\langle v, Q_{\text{pole}} v \rangle = 2L \int_0^1 K_v(\omega) \cosh\left(\frac{L(1-\omega)}{2}\right) d\omega = \int_{-\Delta}^{\Delta} \widehat{g}_v(\xi) 2e^{-\pi\xi} d\xi = 2g_v(i/2),$$

where the middle equality is the substitution $\xi = \Delta(1 - \omega)$ together with $\pi\Delta = L/2$, $\widehat{g}_v(\xi) = \pi K_v(1 - \xi/\Delta)$, and the evenness of $\widehat{g}_v$ (which converts $2 \cosh(\pi\xi)$ into $2e^{-\pi\xi}$ under the integral), and the last equality is $e^{2\pi i(i/2)\xi} = e^{-\pi\xi}$. Applied inside the compact finite- T archimedean integral,

$$\langle v, Q_{\text{arch},T} v \rangle = \frac{1}{2\pi} \int_{-T}^T h_+(r) \left(\int_0^L K_v(1 - y/L) \cos(ry) dy \right) dr = \frac{1}{2\pi} \int_{-T}^T h_+(r) g_v(r) dr,$$

where the inner integral is $g_v(r)$ by the same substitution $\xi = \Delta(1 - y/L)$ and evenness of $\widehat{g}_v$. Lemma 2.2 and $h_+(r) = O(\log(2 + |r|))$ [23] allow $T \rightarrow \infty$ by dominated convergence. Thus the assembly (4) equals the displayed prime, pole, and archimedean source expression.

The Guinand–Weil explicit formula in the normalization of Connes, Bombieri, and Connes–Consani–Moscovici [19, 28, 6, 4, 10] states, for every test function in the class of Lemma 2.2,

$$\sum_{z \in Z_\zeta^*} g_v(z) = -\frac{1}{\pi} \sum_{q=p^a} \frac{\Lambda(q)}{\sqrt{q}} \widehat{g}_v\left(\frac{\log q}{2\pi}\right) + 2g_v(i/2) + \frac{1}{2\pi} \int_{\mathbb{R}} h_+(r) g_v(r) dr.$$

Since $\text{supp } \widehat{g}_v \subset [-\Delta, \Delta] = [-L/(2\pi), L/(2\pi)]$, prime powers $q > c$ satisfy $\log q/(2\pi) > \Delta$ and do not contribute. This identifies the finite form with the zero side. $\square$

Remark 2.6 (What is new here). Once g_v is known to be admissible, the displayed identity is the explicit formula applied to g_v ; in that sense the zero-sum representation is the classical mechanism of Weil [28], in the finite-rank setting studied by Yoshida [29] and built by Connes–van Suijlekom/Connes–Consani–Moscovici. The content of Theorem 2.5 is the exact closed-form transport $v \mapsto g_v$ and its calculus: the truncated matrix is not merely positive-semidefinite-approximating data, but evaluates exact zero sums for an explicitly parametrized $(N+1)$ -dimensional family of band-limited test functions, with the source dependence factoring through the exact $2N+1$ -dimensional quotient of Corollary 2.4. This is the finite, inverse-free part of the correspondence between coefficient vectors and test functions; no claim is made about realizing arbitrary Guinand–Weil test functions this way.

Corollary 2.7 (Pole-neutral source survival). *Let $\beta = L/(4\pi)$ and, for $s \geq 0$, put*

$$H_s(N) = \{v : M_0(v) = M_2(v) = \cdots = M_{2s}(v) = 0\},$$

where

$$M_0(v) = v_0 + \sqrt{2} \sum_{k=1}^N v_k, \quad M_{2j}(v) = \sqrt{2} \sum_{k=1}^N k^{2j} v_k \quad (j \geq 1).$$

Define the pole-neutral hyperplane

$$P_N(c) = \left\{ v : \frac{v_0}{\beta^2} + \sqrt{2} \sum_{k=1}^N \frac{v_k}{k^2 + \beta^2} = 0 \right\}.$$

If $N \geq s + 2$, then

$$\dim(H_s(N) \cap P_N(c)) = N - s - 1 > 0.$$

Every nonzero vector in this space produces a nonzero test function g_v with $g_v(i/2) = 0$, and $g_v = g_w$ implies $v = w$ or $v = -w$. On this family the dictionary of Theorem 2.5 has no pole term.

Proof. Direct evaluation of the pole source (2) at integer frequencies, and of its x -derivative at integer frequencies, gives

$$\psi_0(n) = C_c \frac{n}{n^2 + \beta^2}, \quad \psi'_0(n) = C_c \frac{\beta^2 - n^2}{(n^2 + \beta^2)^2}, \quad C_c = \frac{L(\sqrt{c} + 1/\sqrt{c} - 2)}{2\pi^2} > 0.$$

Both evaluations are needed: off-diagonal entries of Q_{pole} use only the integer values $\psi_0(n)$, but the diagonal entries are $\psi'_0(n)$, and value agreement at integers alone would not determine them (the diagonal involves all Fourier components of the source; see [11, Sec. 4.2]). With both identities, Q_{pole} is exactly C_c times the divided-difference matrix of $f(x) = x/(x^2 + \beta^2)$, including the diagonal $f'(n)$. From the partial fractions $f(x) = \frac{1}{2}((x - i\beta)^{-1} + (x + i\beta)^{-1})$, the divided difference is

$$\frac{f(m) - f(n)}{m - n} = \frac{\beta^2 - mn}{(m^2 + \beta^2)(n^2 + \beta^2)},$$

with the $m = n$ limit $f'(m)$; contracting against the even embedding, and using $\sum_m u_m m/(m^2 + \beta^2) = 0$ by symmetry,

$$\langle v, Q_{\text{pole}} v \rangle = C_c \beta^2 \left(\frac{v_0}{\beta^2} + \sqrt{2} \sum_{k=1}^N \frac{v_k}{k^2 + \beta^2} \right)^2 = 2g_v(i/2).$$

M	$2 \sum_{n \leq M} g_v(\gamma_n)$	raw residual	tail-corrected residual
32	0.049967509...	$-9.1 \cdot 10^{-7}$	$-1.6 \cdot 10^{-7}$
64	0.049968326...	$-8.8 \cdot 10^{-8}$	$-9.4 \cdot 10^{-9}$
128	0.049968406...	$-7.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-9}$
256	0.049968413...	$-6.5 \cdot 10^{-10}$	$-8.3 \cdot 10^{-11}$
512	0.049968414...	$-4.7 \cdot 10^{-11}$	$-5.4 \cdot 10^{-12}$

Table 1: Zero-side partial sums for the worked example ($c = 13$, $N = 4$, pole-neutral v ; γ_n = ordinate of the n -th nontrivial zero, $\gamma_{512} = 826.90\dots$). The limit is the closed-form matrix contraction $\langle v, Q_\infty v \rangle$; residuals are signed.

Thus $P_N(c)$ is exactly the pole-neutral row.

The row defining $P_N(c)$ is independent of the moment rows when $N \geq s + 1$: otherwise there would be a polynomial P of degree at most s with $P(k^2) = 1/(k^2 + \beta^2)$ for $0 \leq k \leq N$; then $(x + \beta^2)P(x) - 1$, of degree at most $s + 1$, would vanish at the $N + 1 \geq s + 2$ distinct points k^2 , forcing it to vanish identically, which is impossible. The dimension formula follows.

Finally, $K_v = 2(T_v * T_v)$ in the Volterra convolution algebra of analytic germs at 0. This algebra is an integral domain: if ϕ, χ have lowest nonvanishing orders j, k , the lowest coefficient of $\phi * \chi$ is a nonzero beta-integral multiple of the product of lowest coefficients. Hence $K_v \equiv 0$ implies $T_v = 0$, so $v = 0$. If $g_v = g_w$, Fourier injectivity gives $K_v = K_w$ on $[0, 1]$, hence as germs; then $(T_v - T_w) * (T_v + T_w) = 0$, so $T_v = \pm T_w$ and $v = \pm w$. $\square$

2.3 A worked example over the first 512 zeros

Take $c = 13$, $N = 4$, and the pole- and moment-neutral vector $v \in H_0(4) \cap P_4(13)$ with $(v_2, v_3, v_4) = (1, 0, -3)/\sqrt{2}$ and (v_0, v_1) determined by the two linear conditions; numerically

$$v = (-0.0859452, 1.4749860, 0.7071068, 0, -2.1213203).$$

The chain $v \mapsto T_v \mapsto K_v \mapsto \hat{g}_v \mapsto g_v$ for this vector is displayed in Figure 1. The cutoff-free contraction, computed from the closed-form Connes–Consani–Moscovici assembly at 40 digits, is

$$\langle v, Q_\infty v \rangle = 0.049968414571096979730\dots,$$

and Theorem 2.5 asserts that this number is the sum of g_v over the nontrivial zeros; since every zero in the range used (and far beyond γ_{512}) is known to lie on the critical line, the zero side is $2 \sum_{n \geq 1} g_v(\gamma_n)$ over the ordinates. Table 1 shows the partial sums against the first 512 zeros: the raw residual falls to $-4.7 \cdot 10^{-11}$, and correcting the remainder by the archimedean tail integral of Section 3 (the smooth zero-density approximation to the missing zeros) leaves $-5.4 \cdot 10^{-12}$. The same package verifies the identity on a generic, non-neutral vector at $(c, N) = (29, 6)$, where the two sides of Lemma 2.1 agree to $2.0 \cdot 10^{-10}$, and on its pole-neutral counterpart, where the pole term vanishes to $9 \cdot 10^{-41}$ while the prime and archimedean terms are of order one: the pole really is switched off, not merely small.

3 Archimedean tail order

Recall $\rho = 2\pi/L$ and $I_N = \{-N, \dots, N\}$. For $T > \rho N$, set $a_T = T/\rho$ and define

$$p_T(n) = \frac{1}{a_T - n}, \quad q_T(n) = \frac{1}{a_T + n} \quad (n \in I_N).$$

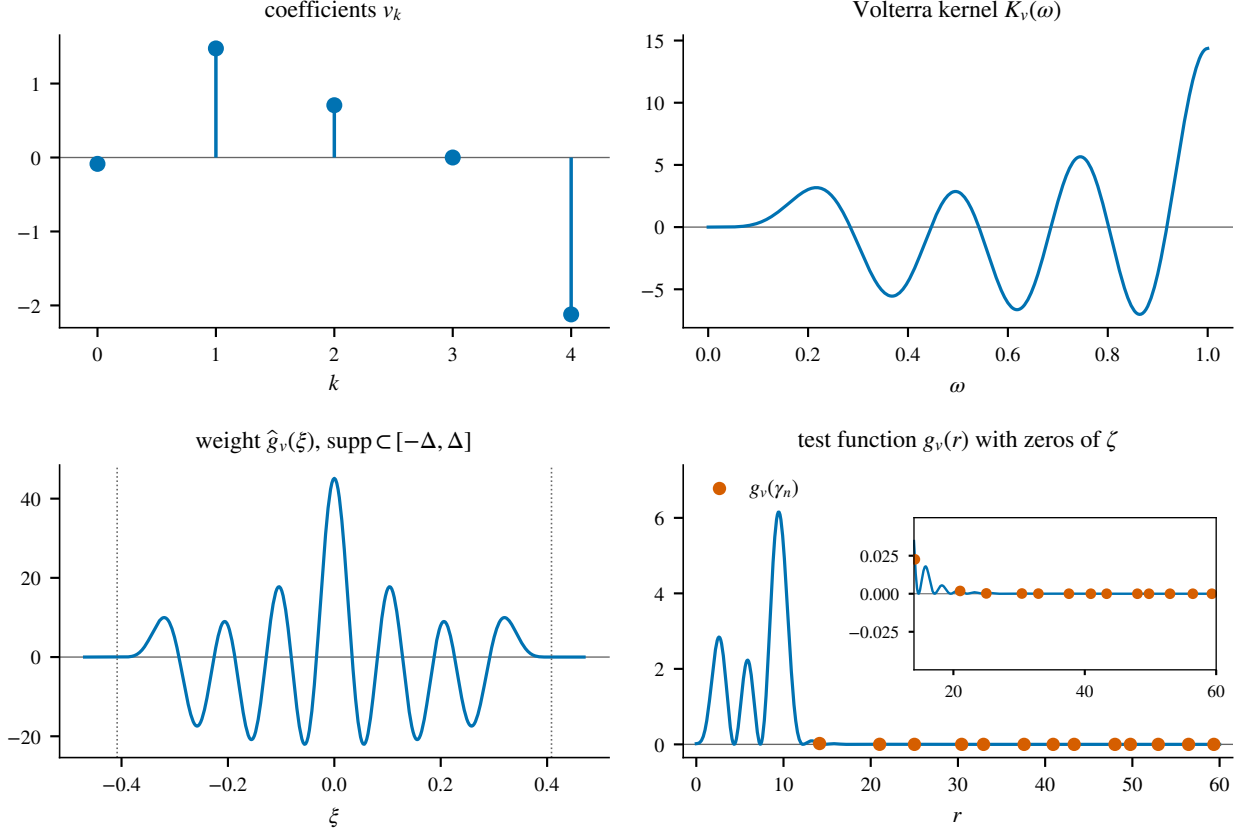


Figure 1: The finite dictionary at $c = 13$, $N = 4$ for the worked example, read in row order. Top row: the coefficient vector v ; the Volterra kernel K_v on $[0, 1]$. Bottom row: the compactly supported Fourier weight $\hat{g}_v$ on $[-\Delta, \Delta]$; the induced entire test function g_v on $[0, 60]$ with the first ordinates γ_n marked (inset: the small tail oscillation that carries the zero sum). By Theorem 2.5 the quadratic value $\langle v, Q_\infty v \rangle$ equals twice the sum of the marked values (continued over all zeros).

Lemma 3.1 (The archimedean density). *The function h_+ is strictly increasing on $(0, \infty)$, with*

$$h'_+(t) = \frac{t}{2} \sum_{k=0}^{\infty} \frac{k + 1/4}{((k + 1/4)^2 + (t/2)^2)^2} \leq \frac{1}{t} + \frac{13}{10t^2} \quad (t > 0).$$

Interval evaluation in Arb [20] gives $h_+(7) = 0.10717967\dots > 0$, so $h_+(t) > 0$ for $t \geq 7$; combined with the derivative bound,

$$h_+(t) \leq \log t - \frac{8}{5} \quad (t \geq 7).$$

Proof. Differentiating the partial-fraction expansion of $\operatorname{Re} \psi_\Gamma(1/4 + it/2)$ term by term gives the displayed series, which is positive for $t > 0$; monotonicity and, with the certified value $h_+(7) > 0$, positivity on $[7, \infty)$ follow. For the upper bound, put $u = t/2$ and $g_u(x) = (x + 1/4)/((x + 1/4)^2 + u^2)^2$. The function $s \mapsto s/(s^2 + u^2)^2$ increases up to $s = u/\sqrt{3}$ and decreases thereafter, so g_u is unimodal in $x \geq 0$ and

$$\sum_{k \geq 0} g_u(k) \leq \int_0^\infty g_u(x) dx + \max_{s > 0} \frac{s}{(s^2 + u^2)^2} = \frac{1}{2(u^2 + 1/16)} + \frac{9}{16\sqrt{3}u^3} \leq \frac{1}{2u^2} + \frac{9}{16\sqrt{3}u^3},$$

whence $h'_+(t) = u \sum_k g_u(k) \leq 1/(2u) + 9/(16\sqrt{3}u^2) = 1/t + (9/(4\sqrt{3}))/t^2 \leq 1/t + 13/(10t^2)$. Integrating from 7 to t ,

$$h_+(t) \leq h_+(7) + \log \frac{t}{7} + \frac{13}{10} \left(\frac{1}{7} - \frac{1}{t} \right) \leq \log t + \left(h_+(7) + \frac{13}{70} - \log 7 \right) \leq \log t - 1.65,$$

using the certified $h_+(7) < 0.1072$. □

Theorem 3.2 (Archimedean tail order). *Let $T_2 > T_1 > \max(\rho N, 7)$. Then*

$$\Delta_{T_1, T_2} := Q_{\text{arch}, T_2} - Q_{\text{arch}, T_1}$$

has the exact representation

$$\Delta_{T_1, T_2} = \frac{1}{\pi^2} \int_{T_1}^{T_2} h_+(T) \frac{\sin^2(LT/2)}{\rho} (p_T p_T^\dagger + q_T q_T^\dagger) dT.$$

Consequently Δ_{T_1, T_2} is positive definite on the full complex finite frequency space indexed by I_N , and hence on the real even subspace. Moreover, in the natural order $-N < \dots < N$, every minor of Δ_{T_1, T_2} with increasing row and column sets is strictly positive.

Proof. Since the finite- T archimedean integrand is even in r ,

$$\frac{d}{dT} \psi_{\text{arch}, T}(x) = \frac{1}{\pi^2} h_+(T) \mathcal{S}(T, x, L).$$

At an integer node $n \in I_N$, direct integration gives

$$\mathcal{S}(T, n, L) = \frac{2\rho n \sin^2(LT/2)}{T^2 - \rho^2 n^2} = \frac{2 \sin^2(LT/2)}{\rho} \frac{n}{a_T^2 - n^2},$$

and for the diagonal entries, differentiating under the integral sign and integrating by parts,

$$\partial_x \mathcal{S}(T, x, L) \Big|_{x=n} = \int_0^L 2\pi \left(1 - \frac{y}{L}\right) \cos\left(2\pi n \left(1 - \frac{y}{L}\right)\right) \cos(Ty) dy = \frac{2 \sin^2(LT/2)}{\rho} \frac{a_T^2 + n^2}{(a_T^2 - n^2)^2},$$

which is exactly the derivative at $x = n$ of the closed form $x \mapsto (2 \sin^2(LT/2)/\rho) x/(a_T^2 - x^2)$. This matters because the diagonal entries of a divided-difference matrix are derivatives of the true source, not of an integer-node surrogate; here the two agree. The divided difference of $x/(a^2 - x^2)$ is

$$\frac{1}{2} \left(\frac{1}{(a-m)(a-n)} + \frac{1}{(a+m)(a+n)} \right),$$

with the same formula in the diagonal limit, so

$$\frac{d}{dT} (Q_{\text{arch}, T})_{mn} = h_+(T) \frac{\sin^2(LT/2)}{\pi^2 \rho} \left(\frac{1}{(a_T - m)(a_T - n)} + \frac{1}{(a_T + m)(a_T + n)} \right),$$

which is the displayed rank-two density.

For positive definiteness, suppose $x \in \mathbb{C}^{I_N}$ and $x^* \Delta_{T_1, T_2} x = 0$. The integrand is a sum of two absolute squares with the strictly positive scalar factor $h_+(T) \sin^2(LT/2)$ off a discrete set, so by continuity

$$\sum_{n \in I_N} \frac{x_n}{a - n} = 0$$

for all $a = T/\rho$ in $(T_1/\rho, T_2/\rho)$. This rational function has simple poles at the nodes $n \in I_N$ unless it vanishes identically; taking residues gives every $x_n = 0$.

It remains to prove the minor statement. Put $A = T_1/\rho$ and $B = T_2/\rho$. After the change of variables $T = \rho a$, the same representation is

$$\Delta_{T_1, T_2}(m, n) = \int_{[-B, -A] \cup [A, B]} \frac{1}{(s - m)(s - n)} d\nu(s),$$

where ν is the push-forward of $\pi^{-2} h_+(\rho a) \sin^2(\pi a) da$ to both $s = a$ and $s = -a$: a nonnegative measure with positive mass on every nonempty open subinterval. For increasing row nodes $x_1 < \dots < x_k$ and column nodes $y_1 < \dots < y_k$ in I_N , Andréief's identity [1, 18] gives

$$\det[\Delta_{T_1, T_2}(x_i, y_j)] = \frac{1}{k!} \int \det \left[\frac{1}{s_\ell - x_i} \right]_{i, \ell=1}^k \det \left[\frac{1}{s_\ell - y_j} \right]_{j, \ell=1}^k \prod_{\ell=1}^k d\nu(s_\ell).$$

On ordered distinct support points $s_1 < \dots < s_k$, the Cauchy determinant formula [21, Ch. 1] gives

$$\det \left[\frac{1}{s_\ell - x_i} \right]_{i, \ell=1}^k = \frac{\prod_{i < j} (x_i - x_j) \prod_{\ell < r} (s_r - s_\ell)}{\prod_{i, \ell} (s_\ell - x_i)}.$$

All nodes x_i lie strictly between the two support intervals, so if ℓ_- of the s_ℓ lie in $[-B, -A]$ the sign of this determinant is $(-1)^{k(k-1)/2 + k\ell_-}$, depending only on k and ℓ_- . The same holds for the y_j determinant with the same s_ℓ , hence the same ℓ_- , so the two signs cancel and the integrand is nonnegative, and positive on a set of positive $\nu^{\otimes k}$ measure. The integral is therefore strictly positive. This is the classical Cauchy total-positivity mechanism [21, 25, 3], applied to the finite archimedean tail representation above. $\square$

Corollary 3.3 (A two-sided finite certification rule). *On the full finite frequency space $\mathbb{C}^{I_N}$, let*

$$Q_T^{\text{tot}} = Q_{\text{prime}}^{(c)} + Q_{\text{pole}} + Q_{\text{arch}, T}, \quad B_T = \frac{1}{\pi^2} \int_T^\infty h_+(r) \frac{\sin^2(Lr/2)}{\rho} (\|p_r\|_2^2 + \|q_r\|_2^2) dr.$$

For $T > \max(\rho N, 7)$:

(i) $Q_\infty - Q_T^{\text{tot}} \succ 0$ and $0 \leq Q_\infty - Q_T^{\text{tot}} \leq B_T I$, hence

$$\lambda_j(Q_T^{\text{tot}}) < \lambda_j(Q_\infty) \leq \lambda_j(Q_T^{\text{tot}}) + B_T \quad (1 \leq j \leq 2N + 1),$$

and the eigenvalues $\lambda_j(Q_T^{\text{tot}})$ increase strictly to $\lambda_j(Q_\infty)$ as $T \uparrow \infty$.

(ii) Decision rule. $\lambda_j(Q_T^{\text{tot}}) \geq 0$ certifies $\lambda_j(Q_\infty) > 0$; $\lambda_j(Q_T^{\text{tot}}) < -B_T$ certifies $\lambda_j(Q_\infty) < 0$; a negative eigenvalue in the band $[-B_T, 0)$ certifies nothing about the cutoff-free sign.

(iii) Explicit budget. For $N \geq 1$,

$$B_T \leq \frac{2(2N+1)\rho}{\pi^2} \left[\frac{\log T}{T - \rho N} + \frac{1}{\rho N} \log \frac{T}{T - \rho N} \right],$$

and, as $T \rightarrow \infty$ with (c, N) fixed,

$$B_T = \frac{(2N+1)\rho}{\pi^2 T} \left(\log \frac{T}{2\pi} + 1 \right) (1 + o(1)).$$

In particular $B_{2T}/B_T \rightarrow \frac{1}{2}$: each doubling of the archimedean cutoff lowers the certification floor by one binary digit.

The corresponding restricted statements hold on the real even subspace.

Proof. (i) Integrate Theorem 3.2 from T to infinity. The improper integral converges entrywise because $p_r(n), q_r(n) = O(r^{-1})$ uniformly for $n \in I_N$ while $h_+(r) = O(\log r)$; strict positivity follows by restricting to any compact window, and the trace of the positive tail is B_T , so the tail is bounded above by $B_T I$. The eigenvalue inequalities and monotonicity follow from Weyl monotonicity. (ii) is immediate from (i). (iii) For the explicit bound, use $\sin^2 \leq 1$, the envelope $h_+(r) \leq \log r$ of Lemma 3.1, and $\|p_r\|_2^2 + \|q_r\|_2^2 \leq 2(2N+1)\rho^2/(r - \rho N)^2$, then integrate

$$\int_T^\infty \frac{\log r}{(r-b)^2} dr = \frac{\log T}{T-b} + \frac{1}{b} \log \frac{T}{T-b} \quad (b = \rho N < T).$$

For the asymptotic form, $\|p_r\|_2^2 + \|q_r\|_2^2 = 2(2N+1)\rho^2 r^{-2}(1 + O(N\rho/r))$, $h_+(r) = \log(r/2\pi) + o(1)$ [23], and the oscillatory part of $\sin^2 = \frac{1}{2} - \frac{1}{2}\cos(Lr)$ contributes $O(T^{-2}\log T)$ after one integration by parts; the remaining smooth integral is $\frac{1}{2} \int_T^\infty \log(r/2\pi) 2(2N+1)\rho r^{-2} dr = (2N+1)\rho(\log(T/2\pi) + 1)/T$ up to the stated relative error. $\square$

The $T = 800$ correction scale. For $c = 100$, $N = 200$, $T = 800$, the released package gives $\rho N = 272.875270768\dots < 800$, a certified dyadic interval budget $B_{800} < 0.897$ (the interval certificate bounds $\sin^2(Lr/2) \leq 1$ and $h_+(r) \leq \log r$, roughly doubling the exact value 0.4203, against which the asymptotic form of (iii) predicts 0.4051; the closed-form bound of (iii) gives the weaker 1.58), and entrywise tail bound $3.18 \cdot 10^{-3}$. A separate cutoff-free Arb interval $LDL^\top$ certificate at the same (c, N) , at 9000 bits, gives $n_+ = 401$ and $n_- = 0$. Thus the finite- T negative eigenvalues that motivated this paper sat deep inside the inconclusive band of the decision rule and were tail artifacts, while the cutoff-free matrix is certified positive. The asymptotic of (iii) makes the obstruction quantitative: driving B_T below a spectral scale of 10^{-59} at $(c, N) = (100, 200)$ would require $T \approx 8 \cdot 10^{62}$. Deep spectral scales are not reachable through the archimedean cutoff; they are reachable through the cutoff-free closed forms.

4 Verification and scope

The released package records exact integer or symbolic audits for the single-frequency identity, the $2N+1$ source quotient through $N = 30$, and the pole-neutral support corollary, including the diagonal derivative identity of Corollary 2.7. The pole-neutral audit checks 3311 pole-square entries, 18 moment-independence determinant cases, and 45 dimension instances of the formula $\dim = N - s - 1$. The archimedean audits include an Arb interval certification of $h_+(7) > 0$, a three-route derivative bridge for the rank-two density (symbolic residual 0; direct-kernel error $9.5 \cdot 10^{-125}$), strict-total-positivity smoke tests with zero bad minors for $N = 2, 3$ (251 and 3431 minors), the $c = 100, N = 200, T = 800$ interval tail budget quoted above, and the cutoff-free Arb $LDL^\top$ certificate with $n_- = 0$ at 9000 bits, whose generating script ships with the package. The dictionary itself is confirmed by three independent routes: the closed-form Connes–Consani–Moscovici assembly, the source-side evaluation of Theorem 2.5, and the zero-side sum of Table 1; on a generic vector at $(c, N) = (29, 6)$ the first two routes agree to $2.0 \cdot 10^{-10}$, and all displayed constants are generated by scripts included in the package. These checks guard constants and signs; they are not proof substitutes.

The limitations are explicit. The dictionary is one-way, from finite Galerkin vectors into the Guinand–Weil class; no inverse map from arbitrary test functions is claimed. Strict observability and strict total positivity concern the isolated post-band archimedean increment, not the full Weil

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